MILITARY SPECIFICATION

RADIO DATA LINK, SOUND RANGING

AN/GRA-114 (SYSTEM OF)

This specification is approved for use by Electronics Research and Development Command, Department of the Army, and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for Radio Data Link, Sound Ranging AN/GRA-114 and testing in conjunction with Sound Ranging Set AN/TNS-10.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The effective issue or revision of the following documents shall be that listed in the Department of Defense Index of Specifications and Standards (DoDDISS) and supplements thereto, unless (i) specific issues are set forth therefor in the cited specifications or (ii) issues different from those specified in the cited specifications are set forth in the solicitation. The date of the applicable DoDISS and supplements thereto shall be as specified in the solicitation or contract.

SPECIFICATIONS

MILITARY

MIL-P-116	- Preservation-Packaging, Methods of.
MIL-R-49111(EL)	- Receiver-Transmitter RT-1183( )/GRA-114.
MIL-R-49112(EL)	- Receiver-Transmitter RT-1184( )/GRA-114.
MIL-R-49113(EL)	- Receiver-Transmitter RT-1185( )/GRA-114.
MIL-C-49114(EL)	- Control, Receiver-Transmitter C-9887( )/GRA-114.
MIL-R-49122(EL)	- Recorder, RO-481/TNS-10.
MIL-M-49123(EL)	- Microphone M-161( )/TNS-10.
MIL-M-55692	- Handset H-250/U.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this documents should be addressed to: Commander, US Army Electronics Research and Development Command, ATTN: DELCS-PE, Fort Monmouth, New Jersey 07703-5304 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1290

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MIL-STD-252	- Classification of Visual and Mechanical Defects for Equipment, Electronic, Wired and Other Devices.
MIL-STD-781	- Reliability Tests Exponential Distribution

DRAWINGS

ES-D-214267	- Microphone Shelter MX-9700/TNS-10.
DLSM-B-493207	- Mast AB-903/G.
DLSM-B-706721	- Antenna AS-4030/GRA-114 with cables.
DLSM-B-706722	- Antenna AS-4031/GRA-114 with Whip-Antenna and cables.
DLSM-B-706723	- Antenna AS-4032/GRA-114 with Whip-Antenna and cables.
DLSM-B-706747	- Filter Amplifier.
DLSM-B-706657	- Kit, Crystal Module, MK-1750/GRA-114
SPI 1G00048	- Special Packaging Instruction.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 System description. The Radio Data Link, Sound Ranging AN/GRA-114 shall be constructed and tested in accordance with the requirements of specifications and drawings listed below and tested as a system in accordance with the requirements of this specification.

Item	Specification or drawing	Quantity
Antenna AS-4030( )/GRA-114	DL SM-B-706721	1
Antenna AS-4031( )/GRA-114	DL SM-B-706722	8
Antenna AS-4032( )/GRA-114	DL SM-B-706723	2
Conversion Kit, Antenna	SM-D-707123	1
Case, Electronic Components CY-7508( )/GRA-114	DL SM-B-706957	1
Control, Receiver-Transmitter C9887( )/GRA-114	MIL-C-49114	2
Crystal Module Kit MK-1750( )/GRA-114	DL SM-B-706657	1
Hammer, hand, 2.5 lbs.	GGG-H-86,Type X, Class 1	1
Handset H-250/U	MIL-H-55692	5
Mast AB-903/G	MIL-HCD-586	1
Receiver-Transmitter RT-1183( )/GRA-114	MIL-R-49111	1
Receiver-Transmitter RT-1184( )/GRA-114	MIL-R-49112	8
Receiver-Transmitter RT-1185( )/GRA-114	MIL-R-49113	2
Technical Manual TM 11-5895-1095-12	Book Plan	2

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3.2 First article. When specified (see 6.2) the contractor shall furnish first article systems for first article inspection and approval (see 4.3).

3.3 Performance requirements. The Radio Data Link, Sound Ranging AN/GRA-114 system shall meet the system performance requirements of 3.3.1 through 3.3.7 after each unit has met the requirements of drawing or specification of 3.1.

3.3.1 Voice communications. The voice communications between stations of the system shall be clear and free of audio distortion (see 4.6.1). It shall be possible to activate the system by commands originating at the SO's or CP and to return to standby Command from the CP.

3.3.2 Command signals. During the command signals operation each position on the code setting switches at the Command Post and SO's shall be checked and the operation of each lamp indicator at the CP shall be verified. (see 4.6.2).

3.3.3 Automatic shutdown. The timing of automatic shutdown of the MP transmitters as indicated from START command to final MP units shutdown (the last lamp indicator shutdown), shall be measured and shall be not more than 6 minutes or less than 4.4 minutes (see 4.6.3).

3.3.4 Data transmissions. Each of the microphone position shall meet the requirements of data transmissions tests of 4.6.4.

a. Audio-signals: 5 HZ, 10 HZ, 25 HZ.
b. Simulated weapons firing.
c. M-161 pumping transient.

3.3.5 Telemetering. During this test, the warning signal for depleted battery condition of a microphone position, shall appear at the Command Post (see 4.6.5).

3.3.6 Phase shift. Phase shift shall not exceed 1.0 millisecond between any two channels from 5-100 Hz. In addition, phase relationships shall correspond within 1.3 milliseconds to signals transmitted over wire lines (see 4.6.6).

3.3.7 Start-up delay. All components required in the transmission and reception of signal data shall be fully operational and free of anomalies within 300 ms after transmission of a data transmit command signal (see 4.6.7).

3.4 Reliability. In lieu of the specified units reliability requirements and tests, the system shall have a specified (0_O) mean-time-between-failure (MTBF) as defined in MIL-STD-781, of 400 hours. The lower test MTBF (0_1) as defined in MIL-STD-781 shall be 200 hours. The reliability requirements shall be demonstrated when operated under the conditions specified in 4.7.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3). (Does not include packaging.)

b. Inspections covered by subsidiary documents (see 4.4).

c. Quality conformance inspections.
Quality conformance inspection of equipment before packaging (see 4.5).
Packaging inspection (see 4.9).

4.3 First article. Unless otherwise specified in the contract, the first article inspection shall be performed by the contractor.

4.3.1 First article units. The contractor shall furnish one to four first article systems of the Radio Data Link, Sound Ranging AN/GRA-114 for group C testing.

4.3.2 First article inspection. The first article inspection shall consist of the inspections specified in table I, and shall be performed in the order specified.

4.3.3 First article data. The first article test plan and test reports shall be as required in the contract.

TABLE I. First article inspection.

Inspections	Reqt paragraph	Test paragraph	Order of tests
1. Inspections covered by subsidiary documents	3.1	4.4	Inspection to be performed on all units
2. Group C inspection	See table II		Tests to be performed on all systems
3. Group D inspection			Test to be performed
Reliability	3.4	4.7	on all systems

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4.4 Inspections covered by subsidiary documents. Each item of 3.1 the following shall be inspected under the applicable documents as part of the inspection required by this specification, and the inspection requirement specified in the contract.

4.5 Quality conformance inspection of equipment before packaging. The contractor shall perform the inspections specified in 4.4, 4.5.1 and 4.5.2. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group C inspection. The inspection shall consist of the inspection specified in table II. Samples shall be selected in accordance with 4.5.1.1.

4.5.1.1 Sampling for group C inspection. One system shall be assembled from units selected at random. The units shall be selected from the first quality conformance inspection lot. This sample system shall constitute the group C requirement for all systems produced. No group C sample required if inspection was performed during first article.

TABLE II. Group C inspection.

Inspection	Requirement paragraph	Test paragraph
Visual and mechanical	3.1	4.8
Voice communications	3.3.1	4.6.1
Command signals	3.3.2	4.6.2
Automatic shutdown	3.3.3	4.6.3
Data transmission	3.3.4	4.6.4
Telemetering	3.3.5	4.6.5
Phase Shift	3.3.6	4.6.6
Start-up delay	3.3.7	4.6.7

4.5.1.2 Group C failures. Action required relative to group C failures shall be as specified in the contract.

4.5.1.3 Reinspection of conforming group C sample units. Unless otherwise specified, sample system units which has been subjected to and passed group C inspection may be accepted on the contract provided all damage is repaired.

4.5.2 Group D inspection. This inspection shall consist of the test specified in table III and shall be performed on systems from lots which have been subjected to and met group C inspections.

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4.5.2.1 Sampling for group D inspection. One to four systems shall be selected at random for group D inspection from every 100 systems or fraction thereof produced. The first samples selected shall be at the start of the contract from the first quality conformance inspection lot.

TABLE III. Group D inspection.

Inspection	Requirement paragraph	Test paragraph
Reliability	3.4	4.7

4.5.2.2 Group D failures. The case of group D failure occurrence, procedures and requirements specified elsewhere in the contract shall be followed.

4.5.2.3 Reinspection of group D sample units. Unless otherwise specified, sample systems which have been subjected to and have passed group D inspection may be accepted on the contract provided all damage is repaired and the units of system retested in accordance with the group A of respective specification.

4.6 Performance tests. The Radio Data Link, Sound Ranging AN/GRA-114 system shall be tested as indicated in the following paragraphs:

4.6.1 Voice communications tests. Voice communications between each of MP and each SO with Command Post shall be tested for clarity and freedom from audio distortion. During the voice transmission from Command Post to MP's or to SO's verify the priority of command signals (START and STOP) over the voice communication. During the voice transmission from Sound Observer to Command Post verify the priority of command signals (START and STOP) over the voice communication. Verify that the appropriate and no other lamp indicators illuminates when receiving command signals, and that they do not respond to signals coming from sources outside the system. All data shall be recorded. (See 3.3.1)

4.6.2 Command signals tests. The command signals tests shall be performed from (see 3.3.2):

a. Command Post
b. Sound Observer No. 1
c. Remote Control (C-9887)
d. Sound Observer No. 2

a. Set the Command Post Code Baseline switches to 555. Set the Code Baseline switches of each microphone position to match the Command Post Baseline number. Send a START command to the MP units using the START button on the Command Post, RT-1183. Observe whether indicator lamp for each MP channel is illuminated and that the signal level meter shows a satisfactory level for each MP channel. Observe whether Recorder, RO-481/TNS-10 begins to run with wind noise

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traces appearing on all eight channels. Send STOP command to MP units. After approximately 5 seconds, verify that all MP units stop transmitting and that all indicator lamps are extinguished and that Recorder RO-481/TNS-10 returns to standby.

b. Set the Code Baseline switches of Sound Observer No. 1 (SO_1) to 555. Using the Data START button on the RT-1185 (SO_1) send a START command to the Command Post. Observe that the system works properly (as described in a.). Send STOP command from the Command Post and verify that system returns to standby.

c. Using the Data START button on the Remote Control (C-9887) connected to the RT-1185 (SO_1) send a START command to the Command Post. Observe that the system works properly (as described in a.). Send STOP command from the Command Post and verify that system returns to standby.

d. Set the Code Baseline switches of Sound Observe No. 2 (SO_2) to 555. Using the Data START button on the RT-1185 (SO_2) send START command to the Command Post. Observe that the system works properly (as described in a.). Send STOP command from the Command Post and verify that system returns to standby.

Repeat (a), (b), (c) and (d) using each of the following code groups. Watch, at all times, for spurious responses, i.e., indicator lights come ON without channel activation.

110	114	101	141	011	411
111	118	121	181	211	811
112					

4.6.3 Automatic shutdown. The timing of automatic shutdown shall be tested as follows: Insure that the Code Baseline switches of all microphone positions are set to the same code number as the Command Post. Using the START button of RT-1183, send a START command to the microphone position units. Observing the indicator lamps of RT-1183, measure the time from the START command to microphone positions shutdown, indicated by indicator lamp shutdown. All data shall be recorded (see 3.3.3).

4.6.4 Data transmissions test. Each microphone position shall be tested for data transmissions to the Command Post as follows (see 3.3.4):

a. Connect equipment at Command Post as shown in Figure 2. Connect equipments at Microphone Positions as shown in Figure 3. Replace Microphone M-161 input at any one channel with audio signal generator as shown in Figure 5. Set Code Baseline switches on all M.P. units to matching code at the Command Post. Feed a 0.3 volt RMS, 5Hz, 10Hz, 25Hz audio signal into the selected RT-1184. Send a Data Start Command from the Command Post. Note whether recorded signals are sinusoidal and that only wind noise appears on other channels. Send stop command and record the data. Repeat above for all channels.

b. Disconnect signal generator used in 4.9.4 a, connect each RT-1184 to its M-161 microphone as shown in Figure 3. Send a Data START Command. Generate an acoustic transient by firing scarecrow cannon in the vicinity of the microphone positions. Observe the recording chart for each channel. Send STOP command. Record results of test.

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c. With the system active, pump the cover of microphone shelter at a microphone position using the palm of your hand. Command Post operator should note the pumping transient on the recorder. Verify that the channel continues to operate by the presence of wind noise after transient ceases. Record the data. Repeat this test for remaining six microphone positions.

4.6.5 Telemetering test. Each of the microphone positions shall be tested for depleted battery condition (see 3.3.5).

a. Install a depleted BA-4386 battery of approximately 10.0V.DC. or connect a variable voltage power source set to 10.0V DC to RT-1184. Send a START command and verify that the appropriate channel indicator lamp on the command Post, RT-1183 blinks ON and OFF indicating low BA-4386 battery condition. Repeat the test for the remaining microphone positions. Replace weak BA-4386 battery with good battery. Record the data.

b. Connect a variable voltage D.C. power source, set to 4.55 volts, to microphone connector socket on RT-1184 (see Figure 6). Send a START Command and verify that the appropriate channel indicator lamp on the Command Post RT-1183 blinks ON and OFF, indicating weak battery condition in M-161 microphone.

4.6.6 Phase shift test. To determine phase shift, use two-trace comparison method using a dual-trace oscilloscope. See Fig 7.

Set time base on oscilloscope to 1 ms/cm for 50 and 100 Hz. Use 10 ms/cm for 5 and 20 Hz. Connect audio signal generator to input terminals of RT-1184 and to the first input channel of oscilloscope. Connect second input channel of oscilloscope to output of channel 1 on RT-1183. Send start command to activate RT-1184. Adjust sensitivity of scope to produce near-vertical rise-times. Measure and record phase difference (time) between corresponding portions of signals on oscilloscope. Perform test on each channel of RT-1183, using mating RT-1184 transmitter. Perform test on each channel using the following audio frequencies 5Hz, 20Hz, 50Hz, 100 Hz. Measure time difference between direct coupled signals to the oscilloscope and those transmitted through AN/GRA-114. The time differences between those signals coming through the RT-1183 and those coming through the wire-coupled input to the oscilloscope shall not differ from each other by more than 1.3 milliseconds (see 3.3.6).

4.6.7 Start-up delay. Connect components as shown in Figure 8. Set time base on oscilloscope to 100 millisec/cm. Set oscilloscope to external trigger. Feed 100 Hz, 0.3 volt signal into RT-1184. See Figure 5. Send start data command from RT-1183. Observe pattern on oscilloscope. Measure time required for incoming signal to stabilize after equipment activation (see 3.3.7).

4.6.8 System construction and deployment test. The system deployment for test shall be as indicated in Figure 1, and shall consist of the following:

a. One Command Post (CP) (see Figure 2).
b. Eight Microphone Positions (MP) (see Figure 3).
c. Two Sound Observers (SO) (see Figure 4).

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The chosen location shall be large enough to permit the distances between units required in Figure 1. The terrain shall provide line-of-sight between the deployment units. The site must permit the installation of the antenna masts and the digging of holes for microphone.

4.6.8.1 Command Post (CP). The Command Post shall consist of the following system elements connected as indicated in Figure 2:

Receiver-Transmitter RT-1183( )/GRA-114
Mast AB-903/G
Filter Amplifier, DL SM-B-706747
Recorder, RO-481/TNS-10
Antenna, AS-4030/GRA-114 with cables
Handset, H-250/U
Battery, BB-248
Field wire

Filter amplifier shall be mounted on the installed mast. The receive and transmit crystal modules shall be installed in Receiver-Transmitter RT-1183/GRA-114.

Receive modules:	SO Command	140.7 MHZ
	SO Voice	140.6 MHZ
	MP-1	138.1 MHZ
	MP-2	138.2 MHZ
	MP-3	140.8 MHZ
	MP-4	140.9 MHZ
	MP-5	141.2 MHZ
	MP-6	141.3 MHZ
	MP-7	141.5 MHZ
	MP-8	141.6 MHZ
Transmit module:		150.2 MHZ

4.6.8.2 Microphone positions (MPs). Seven microphone positions, 50 meters apart, shall be deployed (Figure 1) and connected as indicated in Figure 3. All MP's (RT-1184/GRA-114) units except MP-7 shall use two each BA-4386 batteries. MP-7 shall be powered by two BA-5598/U batteries. The microphone position No. 8 shall be installed with whip antenna and handset H-250/U, at a maximum distance of 200 meters from Command Post. Each of microphone position shall consist of the following equipments connected as indicated in Figure 3:

Receiver- Transmitter, RT-1184( )/GRA-114
Antenna, AS-4031 with cables (except microphone position No. 8)
Microphone, M-161
Microphone Shelter MX-9700
Microphone cable
Receive-transmit antenna cable
Microphone, M-161 attached to Microphone shelter MX-9700 shall be installed in a hole.
Microphone shall not touch the sides of the hole.

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The following receive, transmit crystal modules shall be installed:

In MP-1:	Receive	150.2MHZ
	Transmit	138.1 MHZ
In MP-2:	Receive	150.2 MHZ
	Transmit	138.2 MHZ
In MP-3:	Receive	150.2 MHZ
	Transmit	140.8 MHZ
In MP-4	Receive	150.2 MHZ
	Transmit	140.9 MHZ
In MP-5	Receive	150.2 MHZ
	Transmit	141.2 MHZ
In MP-6	Receive	150.2 MHZ
	Transmit	141.3 MHZ
In MP-7	Receive	150.2 MHZ
	Transmit	141.5 MHZ
In MP-8	Receive	150.2 MHZ
	Transmit	141.6 MHZ

4.6.8.3 Sound Observer (SO). Two Sound Observers stations shall be deployed as indicated in Figure 1. The Sound Observer No. 1 (SO_1) shall consist of:

Receiver-transmitter, RT-1185
Antenna, AS-4032
Control C-9887
Handset, H-250/U

Control C-9887 with Handset H-250/U shall be connected to RT-1185 by 500 meters field wire. Antenna AS-4032 shall be connected by receive-transmit antenna cable. Sound Observer No. 2 (SO_2) shall consist of:

Receiver-transmitter, RT-1185 with Handset, H-250/U and whip antenna.
The following receive, transmit crystal modules shall be installed:

In SO_1:	Receive	150.2 MHZ
	Transmit-Voice	140.6 MHZ
	Transmit command	140.7 MHZ
In SO_2:	Receive	150.2 MHZ
	Transmit-Voice	140.6 MHZ
	Transmit command	140.7 MHZ

4.6.8.4 Equipments required for system tests. The following equipments are required to perform the system tests:

Item	Quantity
Receiver-transmitter RT-1183( )/GRA-114	1
Receiver-transmitter RT-1184( )/GRA-114	8

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Item	Quantity
Receiver-transmitter RT-1185( )/GRA-114	2
Control, Receiver-transmitter C-9887( )/GRA-114	1
Antenna AS-4030 with cables	1
Antenna AS-4031 with cables and whip antenna	7
Antenna AS-4032 with cables and whip antenna	1
Mast AB-903/G	1
Handset H-250/U	4
Filter Amplifier, DL SM-B-706747	1
Microphone M-161	8
Microphone Shelter MX-9700	8
Recorder, RO-481/TNS-10	1
Recorder paper, PH-438	12 rolls
Kit, Crystal Module, MK-1750/GRA-114	1
Battery 12V (Lead-Acid 6TN), BB-248/U	1
Battery, Dry BA-4386	18
Battery, Dry BA-5598/U	2
Battery, Dry Cell BA-30 or equivalent	48
Battery BA-1090 or equivalent	4
Generator Audio Sine Wave	1
Variable Voltage DC power supply	1
Power supply, 110 VAC (If required for instruments	1
Field wire	500 meters
Scarecrow cannon ZON Model 163 or equal	1
Shovel	2
Dual-trace oscilloscope	1

4.6.8.5 Government loaned equipment.

Item	Quantity
Recorder, RO-481/TNS-10	1
Microphone M-161	8
Microphone Shelter MX-9700	8
Recorder Paper, PH-438	12 rolls

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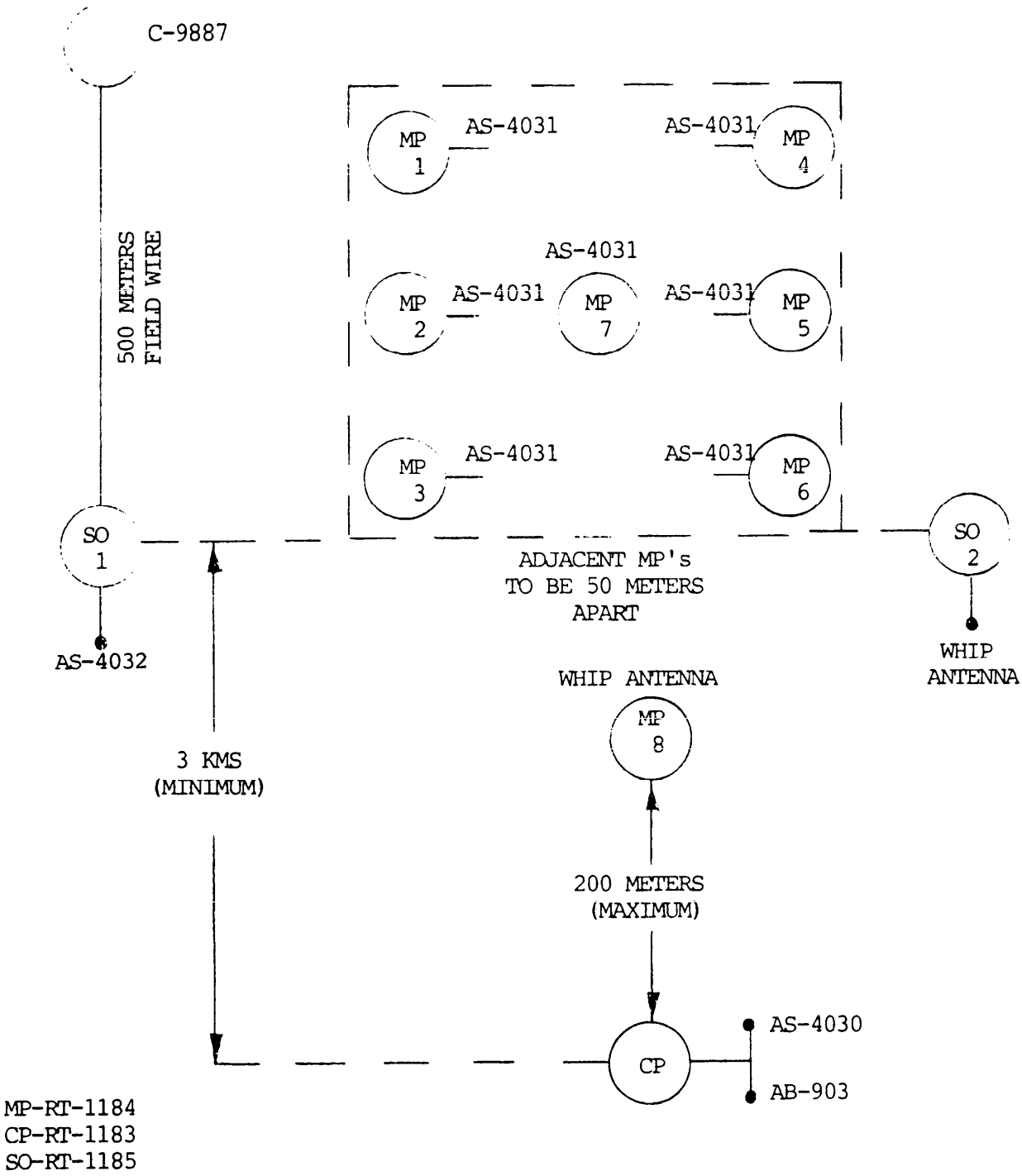


Figure 1. System deployment for test.

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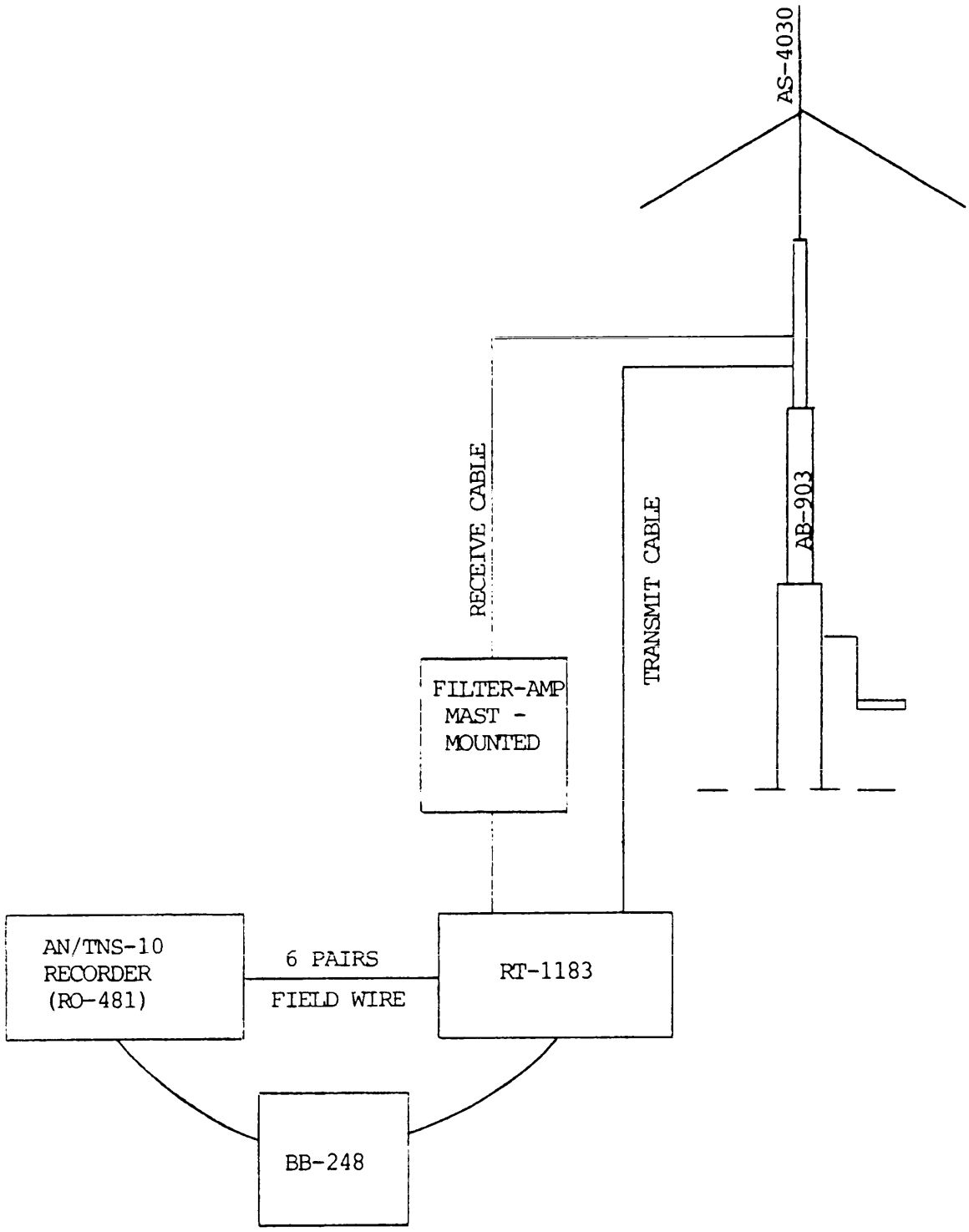


Figure 2, RT-1183 test installation.

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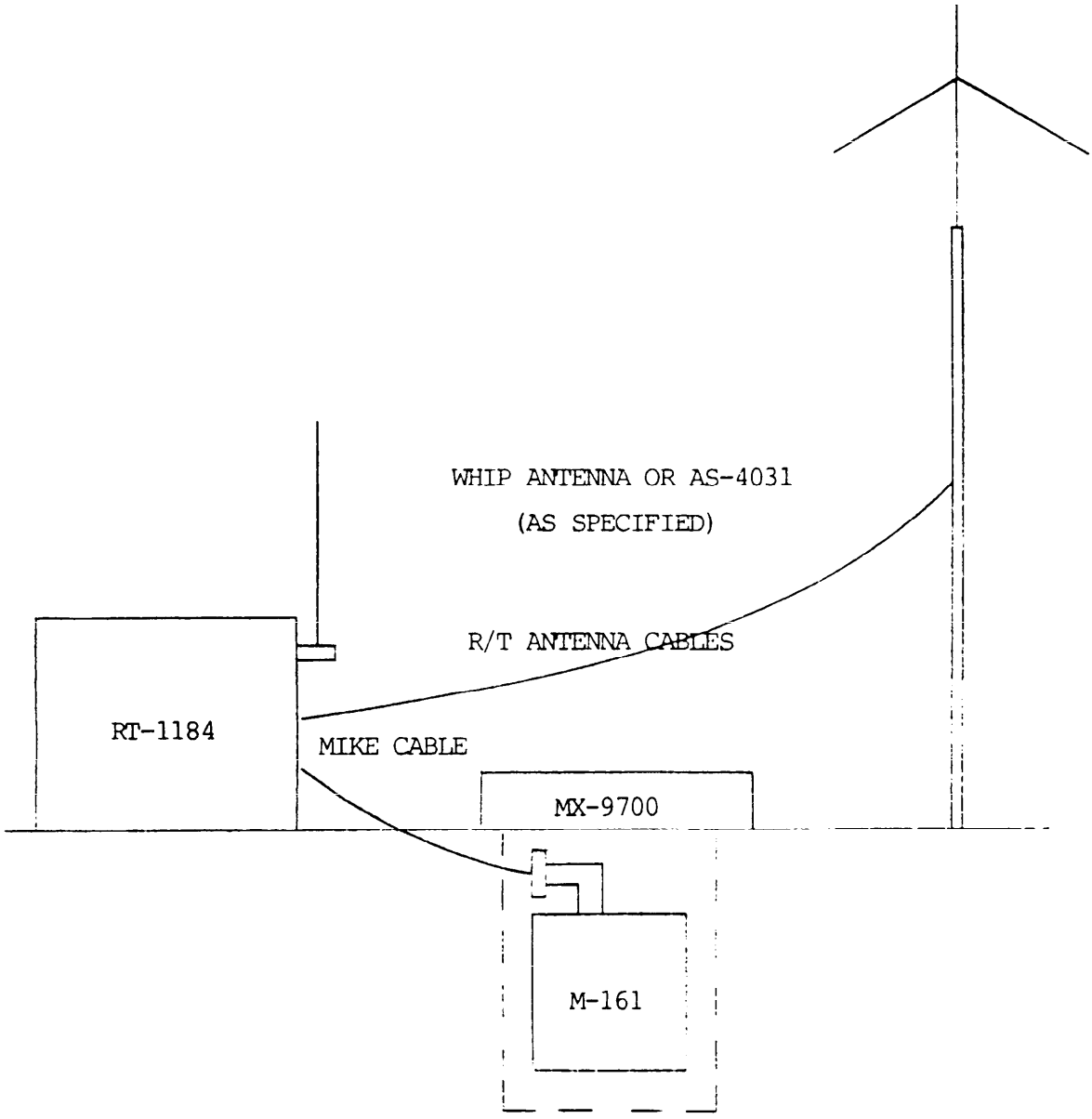


Figure 3. RT-1184(MP) test installation.

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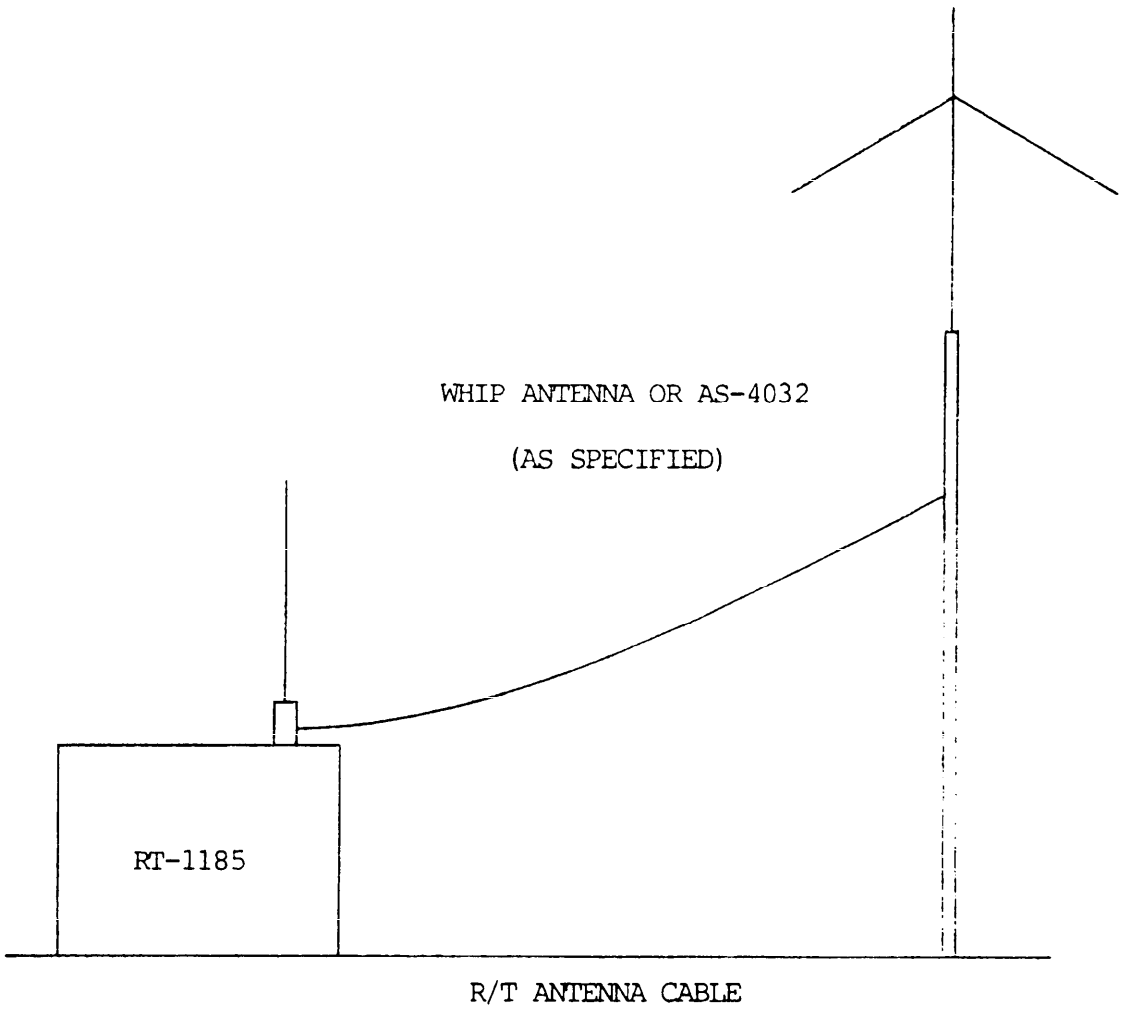


Figure 4. RT-1185 (SO) test installation.

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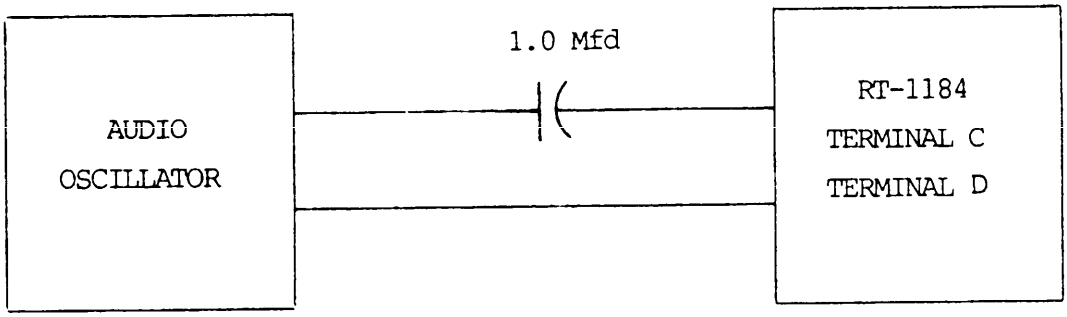


Figure 5. Audio connection.

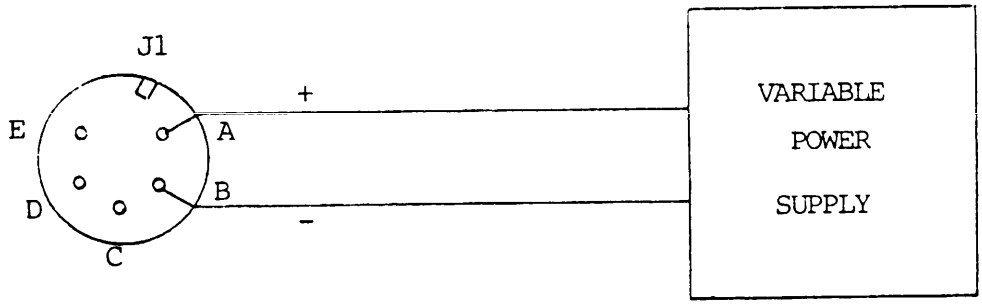


Figure 6. Microphone connector on RT-1184

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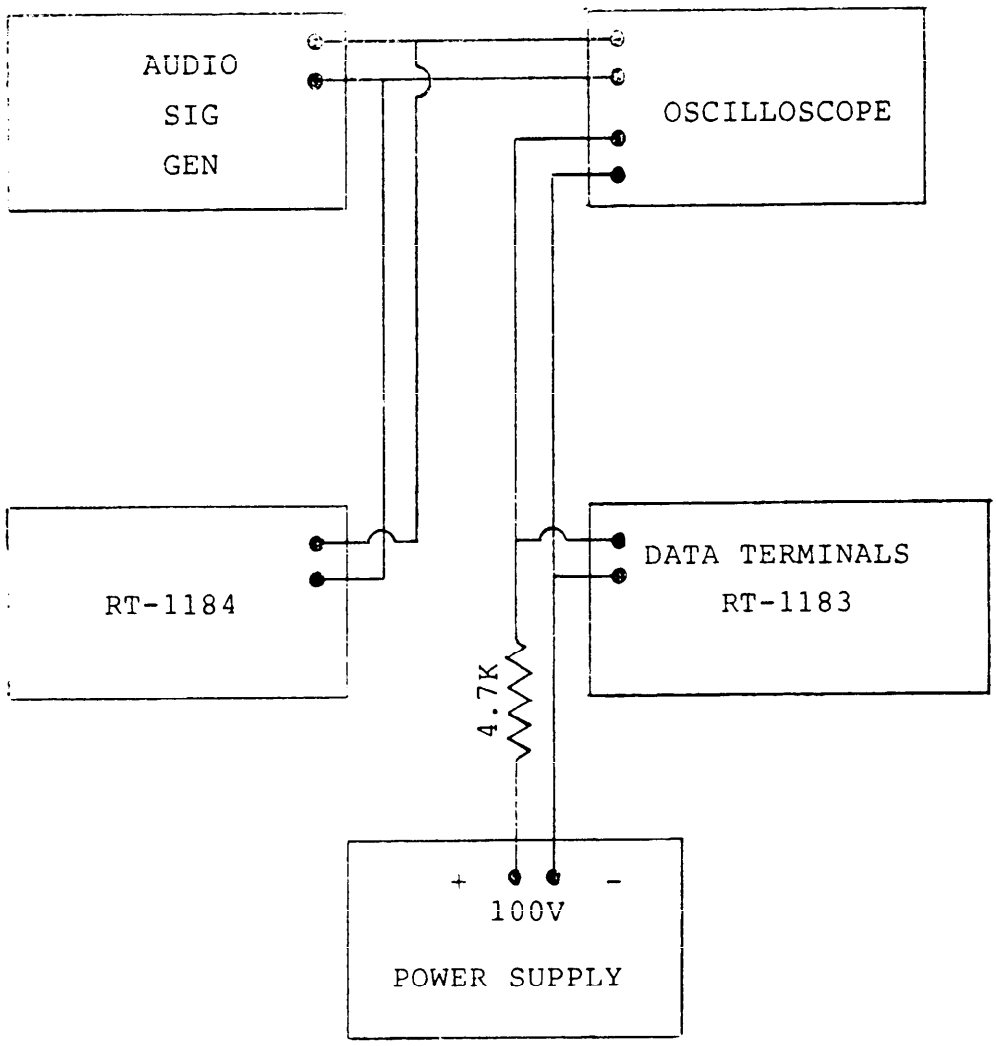


FIGURE 7

Phase shift test.

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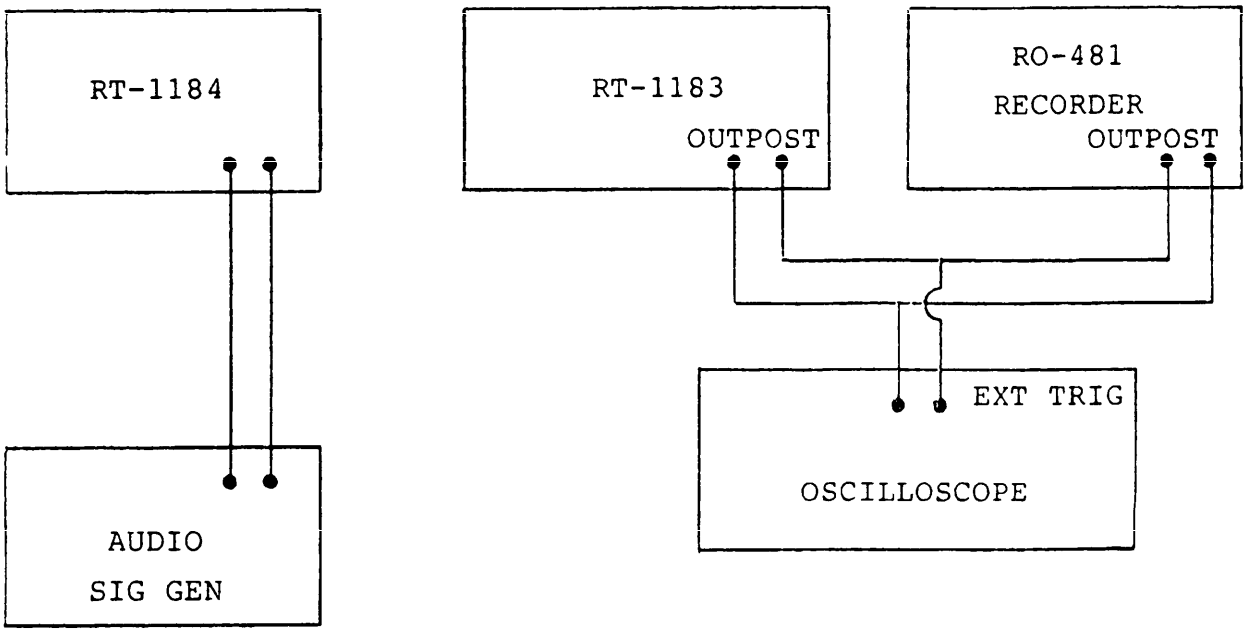


FIGURE 8

Test for start-up delay.

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4.7 Reliability.

4.7.1 Reliability tests. The reliabilitv test shall be conducted in accordance with test plan XIII C of MIL-STD-781. One to four systems shall be used for reliability testing.

4.7.2 Reliability test conditions. Reliability testing shall commence on systems which have successfully completed 4.6.

4.7.2.1 Electrical stress. The input voltage shall be maintained at one hundred ten percent nominal value for one-third of the system ON cycle, at the nominal value for the second one-third of the system ON cycle, and at ninety percent for the final one-third of the system ON cycle. This cycling procedure is to be repeated continuously throughout the reliability test.

4.7.2.2 Thermal stress. The system shall be operated at an ambient temperature of 50°C +5°C, -0°.

4.7.2.3 Test cycle. The system shall be turned ON and let temperature stabilize, hold for 3 hours, then turn OFF and let temperature stabilize. This cycle shall continue throughout the test.

4.7.2.4 Operating stress. All units of the system shall be operated continously in the stand-by mode except for transmitters which shall be operated at full output for approximately 20 seconds each 14.4 minutes of operation.

4.7.3 Failure definition.

4.7.3.1 Unit failure. The inability of a unit to perform its required function is defined a failure. This includes any event which causes departure from performance as specified in the unit specification.

4.7.3.2 System failure. The inability of the system to perform its required function is a system failure. The system is deemed to be unable to perform its required function when any one of the following occur:

a. Any 3 out of the 8 microphone positions fail to perform as specified during any 48 hour operating time period.

b. Both sound observer sets fail to perform as specified during any 48 hour operating time period.

c. Any 3 out of 8 Command Post Data/Voice Receivers fail to perform as specified during any 24 hour operating time period.

d. Both Command Post, Command/Voice Receivers fail to perform as specified during any 24 hour operating time period.

e. The Command Post Transmitter or other Command Post units (excluding the receivers) fail to perform as specified.

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4.7.4 Parameters to be measured.

4.7.4.1 Daily performance parameters. Performance parameters shall be measured and recorded daily in accordance with 5.5.3 of MIL-STD-781. Parameter tests shall be performed in accordance with the procedures given in the equipment specifications listed in paragraph 2.1. The minimum parameters to be measured shall be as follows:

	Parameters	Time interval (hour)
a. RT-1183	Power output	24
	Modulation	48
	Audio distortion	48
	Encode deviation	48
	Command indicator	Post test
	Code receiver sensitivity	24
	Sensitivity	24
	Distortion	48
	Audio output power	48
	Telemetry operation	At least 2 receivers every 24 hours
	Recorder control operation	24
	Frequency	48
b. RT-1184	Power output	24
	Deviation limiting	48
	Telemetry operation	24
	High voltage power supply	Post Test
	Fail safe timer operation	24
	Code sensitivity	24
	Sensitivity	24
	Distortion	48
	Audio output power	48
	Frequency	48
c. RT-1185	Power output-voice	24
	Encode operation	24
	Voice channel frequency accuracy	48
	Remote voice control	48
	Sensitivity	24
	Audio output power	48
d. C-9887	Standby bias	24
	Voice gain bias	24
	Code bias & duration	24
	Line receiver distortion & gain	24
	Voice response (Tx)	48
	Voice response (Rx)	48

4.8 Visual and mechanical. Each unit of Radio Data Link, Sound Ranging AN/GRA-114 shall be examined for the defects listed in MIL-STD-252.

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4.9 Packaging inspection. Packaging inspection requirements specified herein are classified as follows:

a. First article inspection of packaging.
b. Quality conformance inspection of packaging.

4.9.1 First article inspection of packaging. Unless otherwise specified in the contract, first article inspection of packaging shall be in accordance with the Unit Pack Design Validation Requirements of MIL-P-116.

4.9.2 Quality conformance inspection of packaging.

4.9.2.1 Materials inspection. All materials to be used in packaging shall be inspected in accordance with the applicable material specification.

4.9.2.2 Preservation inspection. Inspection of preservation and interior markings shall be in accordance with group A and B Quality Conformance Inspection Requirements of MIL-P-116. Lot formation and sampling procedures shall be as specified therein.

4.9.2.3 Packing inspection. Inspection of packing and the marking for shipment and storage shall consist of the examinations specified in Table IV. Lot formation shall consist of all packs made of the same materials during an identifiable period and submitted at one time for acceptance. Sampling procedures shall be in accordance with MIL-STD-105, using a single sampling plan and acceptable quality level of 4.0 percent defective.

TABLE IV. Packing inspection provision.

No.	Characteristic	Method of inspection
101	Intermediate container not as specified	Visual
102	Improper closure of intermediate container	Visual
103	Shipping containers not in accordance with specification	Visual
104	Excessive cube	Visual
105	Improper blocking and bracing	Visual
106	Closure not in accordance with specification	Visual
107	Weight and size exceed container limitations	Weight & measure
108	Strapping not in accorance with specification, incorrectly applied, omitted	Visual
109	Marking omitted, incorrect, or illegible	Visual

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with SPI 1G00048.

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6. NOTES

6.1 Intended use. The Radio Data Link, Sound Ranging AN/GRA-114 is an FM voice and data link which replaces the field wire used to interconnect the stations of Sound Ranging Set, AN/TNS-10.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification and any amendment thereto.
b. Level A or B preservation and packing (see section 5).
c. When first article inspection rough handling are not required.
d. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.
e. Marking and shipping of samples.
f. Place of final inspection.
g. Number of first articles to be submitted for approval (see 3.1).
h. When first article packaging inspection test reports require Acquisition Activity approval prior to production unit packing.

6.3 Level B preservation. When level B preservation is specified, this level of protection will only be used under known favorable conditions during transportation, storage, and handling.

6.4 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

Custodian:
Army - ER

Preparing activity:
Army - ER
Project 1290-A391

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE

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	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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